

SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY

Big Data analytics (AI358)

Lab Assignment - 9 (R)

Exercise 1: The Air Quality "Mini-ETL"

Goal: Clean a dataset and visualize correlation using a regression line.

- **Dataset:** Built-in airquality dataset. File Name: air_quality_lab.csv
Columns: Ozone (ppb), Solar_R (langley), Wind (mph), Temp (F), Month (1-12), Day (1-31).
- **Tasks:**
 1. Handle missing values (NA) specifically in the Ozone and Solar .R columns.
 2. Use the mutate function to create a "Temperature Category" (Cold, Moderate, Hot).
 3. **Visualization:** Create a scatter plot of Solar .R vs. Ozone. Map the "Temperature Category" to the **color** aesthetic.
 4. **Requirement:** Add a geom_smooth() layer to show the trend.

Exercise 2: Multi-Variable Distribution

Goal: Compare distributions across categories.

- **Dataset:** File Name: server_logs.csv
Columns: Timestamp, Server_ID, CPU_Load (%), RAM_Usage (GB), Response_Time (ms), Error_Count.
- **Tasks:**
 1. Create a **Boxplot** comparing Ozone levels across different Months.
 2. Apply a custom color palette and theme (e.g., theme_minimal()).